Separated degrees and Pell quotients

Smaller explicit consistency equations for ZFC

Research contribution prepared for Vladimir Reshetnikov's ProveIt project

23 October 2026

Abstract

We give several new explicit integer polynomials whose absence of zeros in their stated domains is equivalent, by ordinary mathematics formalizable in ZFC, to syntactic consistency of ZFC. The strongest small-degree result has twelve positive unknowns, degree 242, and a 94-operation division-free evaluation circuit; an 89-operation alternative has degree 258. Separating the ordinary degrees in a packed quadratic source yields a nine-positive-unknown equation of degree 2 628 944 and 499 operations, and a ten-unrestricted-integer equation of degree 8 317 400 and 544 operations. A separate guard construction reduces a fifty-integer quartic to forty-five integer unknowns.

The new ingredients are degree-separated coefficient extraction with an unconditional positive carrier, two separate quadratic norms with exact index recovery, a congruence-based quotient of a Pell ordinate, and shared sign guards. The native computation compiler, machine-to-refutation calibration, and Jones/Sun arithmetic bridges are inherited and explicitly identified. We supply the polynomial circuits, fixed coefficient bytes, mathematical proofs, independent finite audits, and source pins. We claim improvements over the inspected repository frontier, not an absolute minimum, a decision of consistency, or a new end-to-end proof-assistant certificate.

1 Results and the meaning of simplicity

There is no single ordering of Diophantine equations that simultaneously minimizes the number of unknowns, degree, arithmetic circuit size, coefficient description length, and expanded monomial count. The domain is also part of the specification: positive unknowns cannot silently be treated as unrestricted integers. We therefore give explicit improvements in several of these directions, and identify the precise comparisons.

The baseline is the repository revision

`4b4349064a750ba6347293b55874a7f92abdd0db`.

It includes all eight inspected wave-7 archives under `docs/incoming`. The comparison uses those archives as well as the integrated catalogue, rather than comparing only against the older wave-6 summary.

The incoming twelve-positive form has degree 897 and 108 operations. Degree 242 is a 73.02% reduction; the 89-operation variant reduces its evaluator by 17.59%. The best incoming nine-positive degree is 7 344 328, while the smallest incoming operation count at nine positive variables is 548, attained by a different equation. Our equation improves both independent minima: the degree falls by 64.20% and the operation count by 8.94%. Similarly, the incoming ten-integer minima are degree 21 111 400 and 582 operations, again at different points. The new degree falls by 60.60% and the operation count by 6.53%. The quartic result decreases the unrestricted variable count from 50 to 45, a 10% reduction. The detailed archive-by-archive comparison is retained in `proofs/WAVE7_AUDIT.md`.

Domain and unknown count	Exact degree	Operations	Role
$\mathbb{Z}^{12}_{>0}$	**242**	**94**	Lowest-degree new twelve-variable form
$\mathbb{Z}^{12}_{>0}$	**258**	**89**	Shorter twelve-variable evaluator
$\mathbb{Z}^{13}_{>0}$	132	83	Additional tradeoff
$\mathbb{Z}^{9}_{>0}$	**2,628,944**	**499**	Nine positive unknowns
$\mathbb{N}^{9}$	2,628,944	501	Explicit nonnegative translation
$\mathbb{Z}^{10}$	**8,317,400**	**544**	Ten unrestricted integers
$\mathbb{Z}^{45}$	**4**	—	Quartic, 28 quadratic residuals

Table 1: Delivered constructions. Here $\mathbb{N} = \{0, 1, 2, \ldots\}$. Operations are live binary additions, subtractions, and multiplications.

These changes do not improve every existing objective. In particular, the known 31-positive-variable quartic and short fifteen-positive-variable evaluators remain useful points on the frontier. We do not reduce the number nine of positive unknowns or ten of unrestricted unknowns. The forty-five-variable quartic has more expanded symbolic terms than its chosen parent, as discussed in Section 7.

Theorem 1.1 (Mathematical endpoint)**.** *For every polynomial in Table 1, with its domain D and fixed coefficient instance specified in this packet, the supplied reductions and the identified inherited compiler/calibration theorems give an ordinary mathematical proof formalizable in ZFC of*

$$\exists \mathbf{x} \in D\ P(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad \textit{there exists a finite syntactic ZFC refutation.}$$

Consequently ZFC proves

$$\boxed{(\forall \mathbf{x} \in D\ P(\mathbf{x}) \neq 0) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.}$$

The theorem is a composition of explicit new reductions with named inherited theorems. Its mathematical dependency is the accepted K215 consistency calibration, not an assertion that bounded simulations establish nonhalting. The formalization status is narrower: no newly assembled derivation of the displayed equivalence in the repository's object-language `BProv` calculus is supplied. Section 8 explains this distinction.

For a practical choice, the degree-242 twelve-positive form is the most balanced new construction in this packet. If unknowns must range over all integers, use the ten-integer form for a small variable count or the forty-five-integer form for degree four. Every register appearing below is a polynomial abbreviation; only the explicitly listed coordinates are quantified.

2 Explicit coefficients and inherited interfaces

The nine-, ten-, twelve-, and thirteen-variable constructions use one complete coherent four-port tuple

$$(\beta, K_0, M_C, M_F), \qquad \beta = 2^{590069} - 1.$$

Its identifier is

`wang943-checksum67-shared-anchor-length8807-prime581197`.

The bit lengths of its four ports are respectively $(590069, 592884, 590069, 590070)$. All four unsigned little-endian integers are included under `coefficients/coefficients/`, together with their construction and an independent decoder. They occupy 295,388 raw bytes and 9,445 gzip bytes. The fourth port M_F is already the shifted $M_{F,\mathrm{native}} + 2^{590069} - 1$; it must not be shifted a

second time. The archive contains the literal 943-tile input and the literal K215 transition input needed by its builder.

The raw SHA-256 hashes are given in `COEFFICIENT_BINDING.json`. The independent decoder reconstructs all four whole integers, verifies 66,062 native basis identities, and checks the actual coefficient residues needed by the compiler's rank controller. Its 38,663 one-bit negative controls supplement these exact checks. The unbounded compiler proof is included separately; finite checks are not a replacement for it.

The forty-five-integer quartic uses the complete package-34 tuple

`mixed-parity-unrounded-1064921`.

It has a different binding, even though the arithmetic register names are similar. Its four raw payloads occupy 532,479 bytes. Keeping that complete tuple is necessary for its odd-exponent canonical radix. No coefficient ports are spliced between the two instances.

For the degree-separated source below, all channel coefficient recipes are evaluated at the first tuple. A termwise absolute bound gives

$$M = 2^{1182955}, \qquad \kappa = 1182968, \qquad a_S = 2^{\kappa} = 8192M.$$

Every channel coefficient has absolute value strictly below M. The two additional scale coefficients are $2^{9463734}$ and $2^{1182947}$. These scales and every derived channel coefficient leaf are also materialized in `fixed/`, with a signed magnitude format and exact hashes. There are fourteen such nonliteral leaves, requiring 2,512,713 raw bytes or 47,720 gzip bytes. This byte count is separate from arithmetic circuit complexity.

The inherited interfaces are the positive quadratic native compiler, the normalized Pell compiler and its signed auxiliary-index lemma, the accepted K215-to-refutation calibration, the Jones nine-variable ratio bridge, and the Sun ten-integer bridge. The relevant mathematical proof texts and Jones Lean source are retained under `upstream/`, `low_degree/upstream/`, and `quartic/upstream/`. Their source revision and byte hashes are recorded. These imports are part of the proof, not hidden assumptions about the existence of a solution.

3 Two separate norms and a balanced Pell quotient

The direct positive-variable route admits a substantial reduction that is independent of the nine-variable packing construction. The recommended polynomial has twelve positive unknowns, exact degree 242, and 94 binary operations. A cheaper companion has degree 258 and 89 operations. The incoming wave-7 comparison is twelve positive unknowns, degree 897, and 108 operations. A further thirteen-positive companion has degree 132 and 83 operations. The twelve-positive degree-314, 91-operation intermediate is retained because its reconstruction makes the mechanism especially transparent.

The new mechanism keeps two quadratic norms separate. One combines the main Pell root and the first Pell root while absorbing the index quotient. The other removes the strong Pell root. A known residue of the auxiliary Pell ordinate then permits an exact polynomial factor to be removed, with its size chosen to balance the two remaining terms. All coefficients are the complete compact tuple in the coefficient appendix, with the already shifted high mask. There are no additional existential coefficient ports.

3.1 The literal twelve-variable polynomial

Supply exactly

$$J, F, \alpha, z, i, T, s, w, \eta, \zeta, y, t \in \mathbb{Z}_{>0}.$$

Every other letter in this section denotes a polynomial abbreviation. Set

$$\begin{aligned} q &= \beta J + 1, & X &= wq, & Y &= sq^3, & E &= XY, \\ k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, & H &= 4a + 3, \\ \Delta &= a^2 + H, & \ell &= EkY, & A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2, \\ b_0 &= X + ac, & U &= 4q^4, & r &= k - R - 1, & S &= UH. \end{aligned}$$

Here the transport expressions are

$$\begin{aligned} R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\ N_t &= (K_0 + w)(q - F - \alpha) + q - F - z(q-1). \end{aligned}$$

Define the first norm by

$$\begin{aligned} W &= Ub_0 + H\big(U(t + \ell) + r\big), \\ \mathcal{N} &= (W^2 + U^2G - S^2A)^2 - 4W^2U^2G. \end{aligned} \tag{1}$$

For the second norm put

$$\begin{aligned} I &= ic^2, & D &= 1 + \Delta I^2, & Q &= (\Delta I)^2, & d &= Q - 1, \\ v &= cT, & b &= c + RD. \end{aligned}$$

Choose one of the following two pairs of polynomial abbreviations:

$$(M, C) = ((\Delta ic)^2, c^2) \quad \text{or} \quad (M, C) = (Q, 1).$$

In both cases $Q = MC$. Finally set

$$\begin{aligned} Z &= C(Dv^2 + b^2 - 1) + dy(2 - My), \\ \mathcal{J}_M &= Z^2 - 4C^2Dv^2b^2, \\ \boxed{P_M = N_t(1 + \mathcal{N})(1 + \mathcal{J}_M) - 1 = 0.} & \end{aligned} \tag{2}$$

The first choice is the degree-242 equation; the second is the 258-degree companion. Both are ordinary integer polynomials. The balanced evaluator computes $M = (\Delta I)(\Delta i)$ and $Q = Mc^2$, which is the same polynomial definition with one multiplication saved.

Theorem 3.1. *For each of the two fixed polynomials in (2), ZFC proves that it has no solution in twelve strictly positive integers if and only if* Con(ZFC)*. Their exact degree/operation pairs are* $(242, 94)$ *and* $(258, 89)$*, respectively.*

The theorem composes the reduction below with the inherited native-window compiler and K215 syntactic-refutation calibration. The complete new argument is supplied here; the source texts for the signed Pell recovery, canonical compiler witnesses, and auxiliary congruences are retained in `low_degree/upstream/`, with exact pins. This is an ordinary mathematical proof formalizable in ZFC, rather than a new exported object-language derivation or a newly checked end-to-end proof-assistant artifact.

3.2 Units, positivity, and recovery of the strong root

Both $\mathcal{N}$ and $\mathcal{J}_M$ are squares modulo 4. If (2) holds, its three integer factors are units. The value -1 for either $1 + \mathcal{N}$ or $1 + \mathcal{J}_M$ would force the corresponding norm to equal -2, impossible modulo 4. Thus

$$N_t = 1, \qquad \mathcal{N} = \mathcal{J}_M = 0. \tag{3}$$

This obtains the transport interface before any Pell index argument. Its inherited bounds are

$$\begin{aligned} q \ge 16, \quad C_w = q - F - \alpha > 0, \quad 3q < R < q^4 - 2, \\ X \ge q, \quad Y \ge q^3, \quad E \ge q^4 > R + 2, \qquad k \ge 2, \quad kY < c < k(Y+1). \end{aligned} \tag{4}$$

For example, $C_w \le 0$ would imply $N_t \le q - F - z(q-1) = 1 - F - (z-1)(q-1) \le 0$. The remaining bounds are the authentic whole-tuple mask bounds. Consequently M, C, v, b are positive, $Q > 1$, and $M \ge 2$. All denominators used next are nonzero.

From $\mathcal{J}_M = 0$ define the rational number

$$f = \frac{Z}{2Cvb}.$$

It satisfies $f^2 = D$, so it is an integer. The sign of this recovered integer is not prescribed. Put $y_{\mathrm{old}} = My - 1 > 0$. The exact polynomial identity

$$MZ = Q(Dv^2 + b^2) - 1 - d(My - 1)^2 \tag{5}$$

then gives

$$f^2 - \Delta I^2 = 1, \qquad dy_{\mathrm{old}}^2 = Q(vf - b)^2 - 1, \qquad vf - b = c(Tf - 1) - Rf^2. \tag{6}$$

These are the normalized strong and auxiliary Pell equations. The signed auxiliary theorem uses $|f|$ as its modulus and covers a negative f. To match the preserved ordinary-strong statement explicitly, take $i_{\mathrm{ordinary}} = \Delta i > 0$; then $(i_{\mathrm{ordinary}} c^2)^2 = \Delta(f^2 - 1) = Q$. No positive sign for the reconstructed f is silently assumed.

3.3 A two-root carrier restores the main square

Write $g_0 = \sqrt{G} > 0$ and $\tau_0 = \sqrt{A} > 0$. There is the exact real factorization

$$\mathcal{N} = \prod_{\sigma,\epsilon \in \{-1,1\}} (W - \sigma U g_0 - \epsilon S \tau_0). \tag{7}$$

Since $r = k - R - 1 \ge 1 - R$, $U > R$, and $t \ge 1$, we have $t + r/U > 0$ and

$$W = Ub_0 + S(\ell + t + r/U) > 0.$$

A vanishing factor supplies real roots $g^2 = G$, $\tau^2 = A$ with $W = Ug + S\tau$. Squaring isolates

$$2WUg = W^2 + U^2 G - S^2 A.$$

Thus g is rational, hence integral; then τ is integral as well. This argument remains valid when the two radicands have the same square class.

The sign $g < 0$ is impossible. Indeed

$$\tau_0 < \ell + k, \qquad \frac{b_0 + g_0}{H} > \frac{2ac}{4a+3} > \frac{c}{3} > k,$$

since $a > 1$ and $Y > 3$. For $g = -g_0$ the factor equation would give

$$\tau = \ell + t + r/U + (b_0 + g_0)/H > \ell + k > \tau_0,$$

a contradiction. Therefore $g = g_0 > 0$. Reducing the carrier identity modulo U proves

$$U \mid H(k - R - 1). \tag{8}$$

We have not yet canceled H or recovered the omitted index quotient.

3.4 Recover the exact index before the deleted quotients

Put $A_0 = a+2$ and $P_0 = 2EY+1$. Standard Pell classification gives positive indices p, m, n with

$$\begin{aligned} g &= \chi_{A_0}(p), & c &= \psi_{A_0}(p), \\ |f| &= \chi_{A_0}(m), & I &= \psi_{A_0}(m), \\ |\tau| &= \chi_{P_0}(n), & k &= 2\psi_{P_0}(n). \end{aligned}$$

The first-Pell discriminant is $EY(EY+1)$, with fundamental integer-coordinate unit $(P_0, 2)$. Because $P_0 > A_0$ and $c > k$, monotonicity gives $n < p$. Also $c > 1$ implies $p \geq 2$, whence $c \geq 2A_0 > 2R$, and Pell growth gives $c > 2p$.

Strong divisibility and $c^2 \mid I$ give $p \mid m$. Write $m = pj$. Then

$$I = c\psi_g(j), \qquad \psi_g(j) \equiv jg^{j-1} \pmod{c}.$$

Since $g^2 \equiv 1 \pmod{c}$, g is invertible modulo c, so $c \mid j$. Consequently $c \mid m$ and $m \geq pc > 2p$.

For clarity, the signed auxiliary-index recovery used at this point does not depend on either deleted quotient. Put $M_0 = \Delta I$ and $V = c(Tf-1) - Rf^2$. The second equation in (6) is

$$M_0^2 V^2 - (M_0^2 - 1)y_{\text{old}}^2 = 1.$$

Its Pell index j_0 is odd because its abscissa is divisible by M_0. For the integer odd-index polynomials Q_j,

$$\begin{aligned} \chi_{M_0}(2j+1) &= M_0 Q_j(M_0^2), \\ Q_j(1-A_0^2) &= (-1)^j \psi_{A_0}(2j+1), \\ Q_j(0) &= (-1)^j(2j+1). \end{aligned}$$

Modulo $|f|$, $M_0^2 \equiv 1 - A_0^2$ and $V \equiv -c$. Squaring and doubling, and reducing the signed abscissa index using $0 < 2p < m$, yields $j_0 \equiv \pm p \pmod{m}$, hence modulo c. Modulo c, $M_0^2 \equiv 0$, $f^2 \equiv 1$, and $V \equiv -R$, so $j_0 \equiv \pm R \pmod{c}$. Since $0 < p, R < c/2$, it follows that

$$p = R. \tag{9}$$

This is the signed index argument reproduced in the preserved sources; its inputs have all been verified before using h or γ.

Let $\mu = U/\gcd(U, H)$. Since H is odd and $H \equiv 3 \pmod{q}$, the only possible common prime of q and H is 3. If $3 \mid q$, then $27 \mid Y$ and $H \equiv 3 \pmod{27}$, so the 3-adic valuation is one. Therefore

$$\gcd(U, H) \leq 3, \qquad \mu \geq 4q^4/3 > R.$$

The first Pell recurrence gives $k \equiv 2n \pmod{4EY}$. The modulus $4EY = 4ws^2q^7$ is a multiple of U. Thus (8) gives $2n \equiv R+1 \pmod{\mu}$. But $1 \leq n < R$ implies $|2n - R - 1| < R < \mu$. Hence

$$2n = R+1, \qquad r = k - R - 1 \equiv 0 \pmod{4EY}.$$

In particular $h = r/E$ and r/U are integers. Moreover $n \geq 25$ and $\psi_{P_0}(n) > n$ make $h > 0$. Dividing the carrier by U gives

$$g = b_0 + H(t + \ell + r/U - \tau),$$

so $\gamma = (g - b_0)/H$ is integral. Since $n \geq 2$ gives $k \geq 4P_0 > X$, we have $c > X$; since $\Delta > (a+1)^2$, $g - b_0 > c - X > 0$. Thus $\gamma > 0$.

All normalized compiler equations have now been recovered, including $k - hE - R = 1$ and the positive main quotient. Their signed auxiliary soundness theorem gives a blank K215 halt, hence a finite syntactic ZFC refutation. The proof does not require making the reconstructed f positive on the same assignment.

3.5 Canonical completeness and the removed Pell residue

Start with a canonical positive normalized compiler witness. It has $p = R = 2n - 1$, $h > 0$, and $\gamma > 0$. Its first Pell recurrence gives $U \mid r$. Define

$$t = \gamma + \tau - \ell - r/U. \tag{10}$$

This is a positive integer: since $\tau^2 = \ell^2 + \ell k + 1$ and $\ell > k$,

$$\tau - \ell = \frac{\ell k + 1}{\tau + \ell} > \frac{\ell k}{2\ell + k} > \frac{k}{3} > \frac{r}{U}.$$

The carrier identity follows, so $\mathcal{N} = 0$.

For the new ordinate quotient, one may replace the canonical auxiliary witnesses explicitly. The authentic population decoding gives $R \equiv 3 \pmod 4$. To identify the precise inherited fact, the native radix is divisible by 4, $q = B^N$, $J \equiv 1 \pmod 4$, and $M_C \equiv 2 \pmod 4$. The low mask forces $C_w \equiv 1 \pmod 4$; the formula for R gives $R \equiv C_w + 2 \equiv 3 \pmod 4$. This argument is preserved in the wave-7 squared-modulus source and applies to the whole compact tuple here.

Keep the outer coordinates and the original positive f, i. Put $m_0 = \Delta I$, so $Q = m_0^2$, and choose

$$V = \chi_{m_0}(R)/m_0, \qquad y_{\text{old}} = \psi_{m_0}(R). \tag{11}$$

The odd-index polynomial makes V a positive integer. Its identities give

$$V \equiv -c \pmod{f^2}, \qquad V \equiv -R \pmod c, \qquad y_{\text{old}} \equiv -1 \pmod Q. \tag{12}$$

For the first use $m_0^2 \equiv 1 - A_0^2 \pmod{f^2}$ and $(R-1)/2$ odd; for the second use $c \mid m_0$ and the constant term of $Q_{(R-1)/2}$. For the third, $\psi_x(R)$ is an even polynomial in x, with constant term $(-1)^{(R-1)/2} = -1$. The recurrence $\psi_x(j+1) = 2x\psi_x(j) - \psi_x(j-1)$ proves this assertion by induction.

Because $\gcd(c, f) = 1$ and $f^2 \equiv 1 \pmod c$, the numerator below is divisible by cf (in fact by cf^2). Thus

$$T = \frac{V + c + Rf^2}{cf} > 0$$

is an integer, and $V = c(Tf - 1) - Rf^2$. The Pell identity gives the auxiliary equation. This replaces only T and the auxiliary ordinate; it leaves (10) unchanged. No assumption of cofinal auxiliary witnesses is needed.

For either selected $M \mid Q$, define

$$y = (y_{\text{old}} + 1)/M \in \mathbb{Z}_{>0}.$$

Identity (5) gives $\mathcal{J}_M = 0$. All twelve supplied coordinates are positive, proving completeness. Negating the existence equivalence proves the stated consistency equivalence in ZFC.

3.6 Exact degrees and the intermediate companions

Write the leading homogeneous forms as

$$q_0 = \beta J, \quad X_0 = wq_0, \quad Y_0 = sq_0^3, \quad E_0 = X_0Y_0, \quad k_0 = \eta + \zeta, \quad c_0 = k_0Y_0, \quad \ell_0 = E_0c_0,$$
$$H_0 = 4E_0, \quad U_0 = 4q_0^4, \quad I_0 = ic_0^2, \quad R_0 = q_0^3(q_0 - F), \quad (N_t)_0 = w(q_0 - F - \alpha) - zq_0.$$

The large apparent degree of (1) cancels by an exact polynomial identity. Set

$$D_0 = b_0^2 - G = X^2 + 2Xac - Hc^2 - 1, \qquad Z_0 = Ut + r,$$
$$C_2 = U^2D_0 + 2UHb_0Z_0 + H^2(2U\ell Z_0 + Z_0^2 - U^2\ell k - U^2).$$

Then

$$\begin{aligned}\mathcal{N} = 4U^4H^2\ell^2D_0 + 4U^2Hb_0\ell C_2 + C_2^2 \\ - 4U^4H^2G(\ell k + 1).\end{aligned} \tag{13}$$

Here $\deg D_0 \le 16$ and $\deg C_2 \le 32$. The four summands have respective degree bounds $66, 68, 64, 62$. Only the second reaches 68, with nonzero highest part

$$\mathcal{N}_{68} = 2^{16}w^6s^9k_0^3q_0^{49}(2t - k_0). \tag{14}$$

This proves the upper and lower degree statements by polynomial identities, without treating a sampled diagonal degree as an upper bound.

For the full-modulus choice $M = Q$, the two summands defining Z have maximum degrees 76 and 94. The latter has highest part $-Q_0^2y^2$, where $Q_0 = E_0^4I_0^2$. Thus $\mathcal{J}_M$ has degree 188, and P_M has degree $68 + 188 + 2 = 258$, with highest part

$$\mathcal{N}_{68}Q_0^4y^4(N_t)_0 \neq 0.$$

For the balanced choice, $\deg M = 36$ and $\deg C = 10$. The first summand of Z has degree 86 and highest part $c_0^2R_0^2E_0^4I_0^4$; the second has degree 84. The subtracted term of $\mathcal{J}_M$ has degree 142. Hence $\deg \mathcal{J}_M = 172$, and the highest part of P_M is

$$\mathcal{N}_{68}c_0^4R_0^4E_0^8I_0^8(N_t)_0 \neq 0,$$

of degree 242.

The degree-314 intermediate uses the original ordinate directly:

$$\begin{aligned} e &= Q(Dv^2 + b^2) - 1, & B_{\text{aux}} &= 2Qvb, \\ \mathcal{J} &= (e - dy^2)^2 - B_{\text{aux}}^2D, & P_{314} &= N_t(1 + \mathcal{N})(1 + \mathcal{J}) - 1. \end{aligned}$$

It reconstructs $f = (e - dy^2)/B_{\text{aux}}$, with no canonical residue restriction. Its inner norm has degree 244 and highest part $E_0^{16}I_0^{12}R_0^4$, proving total degree 314.

Finally, retaining $f > 0$ gives the thirteen-positive companion

$$\begin{aligned} N_s &= f^2 - \Delta I^2, & V &= c(Tf - 1) - Rf^2, \\ N_a &= \Delta(f^2 - 1)(V^2 - y^2) + y^2, & P_{132} &= N_t(1 + \mathcal{N})N_sN_a - 1. \end{aligned}$$

A zero forces all factors to be integer units and $\mathcal{N} = 0$. The elementary negative-Pell obstruction

$$x^2 - (m^2 - 1)y^2 \neq -1 \qquad (m \ge 2)$$

first rules out $N_s = -1$. Then $\Delta(f^2 - 1) = (\Delta I)^2$ is a square, so the same obstruction rules out $N_a = -1$. Thus $N_t = N_s = N_a = 1$. For the obstruction, a hypothetical nonnegative solution has $(m - 1)y < x < my$; multiplying by $m - \sqrt{m^2 - 1}$ gives a smaller positive ordinate $my - x$, contradicting descent. The remaining proof is the same carrier/index reconstruction above. Degrees of the factors are $2, 68, 34, 28$, and their nonzero leading product is

$$-\mathcal{N}_{68}E_0^4I_0^2c_0^2T^2f^4(N_t)_0,$$

so the exact degree is 132.

The literal DAGs have 94, 89, 91, and 83 live binary gates for $P_{242}, P_{258}, P_{314}, P_{132}$, respectively. Independent finite checks include full polynomial arrays over three primes using the actual fixed coefficients, four-factor multiplication in the formal quadratic algebra, unit-guard congruences, the exact quotient identity, and nontrivial canonical Pell reconstructions. These validate the delivered arithmetic; the universal proof and the identified inherited compiler arguments establish the theorem.

4 One packed word, three ordinary-degree channels

The incoming selective encodings improve on full Sidon separation by isolating only monomials that actually occur in the parent system. Their homogenized packed square still mixes constant, linear, and quadratic source terms. The new observation is that these three ordinary degrees can be evaluated separately without introducing another unknown.

Let $r_1, \dots, r_{22}$ be the raw quadratic residuals of the 31-positive-variable parent in `data/parent_quartic31.json`. The raw positive coordinates are retained, rather than expanding a positive-to-nonnegative shift into the source coefficients. In their specified order, use the weights

$$(w_1, \dots, w_{31}) = (104, 76, 79, 60, 108, 8, 99, 55, 36, 54, 98, 11, 7, 10, 44, 32,\\ 106, 22, 17, 1, 3, 4, 37, 89, 97, 107, 39, 40, 86, 45, 92),$$

and, in residual order, the target positions

$$(q_1, \dots, q_{22}) = (1081, 1388, 1366, 763, 850, 546, 1414, 1147, 1914, 1093, 821,\\ 1861, 39, 1621, 238, 759, 500, 1711, 1052, 283, 1343, 368).$$

Put

$$W = 108, \quad Q_c = 1914, \quad K_c = Q_c + 2W + 1 = 2131, \quad E_c = K_c + W + 2 = 2241,$$
$$H_c = E_c + W + 23 = 2372.$$

The 31-coordinate order is exactly

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).$$

For a source coefficient $p_{j,\alpha}$, write $d = |\alpha|$. Place its reversed term at exponent $q_j - \alpha \cdot w$ in $A_d(T)$. Its coefficient is $2p_{j,\alpha}$ for a constant, a linear monomial, or a square; for a product of two different coordinates it is $p_{j,\alpha}$. Appendix A gives every coefficient and exponent. There are 45 quadratic records, 41 linear records, and eight constant records.

Lemma 4.1 (Degree-separated extraction)**.** *For indeterminates $z_1, \dots, z_{31}$, define*

$$Y_z(T) = \sum_i z_i T^{w_i}, \qquad \mathcal{F}(T, y) = A_2(T)y^2 + A_1(T)y + A_0(T) + T^{Q_c+1}(y^2 + 1).$$

Then, for every $1 \le j \le 22$,

$$[T^{q_j}]\mathcal{F}(T, Y_z(T)) = 2r_j(z_1, \dots, z_{31}).$$

Proof. The finite certificate checks, separately for each $d \in \{0, 1, 2\}$, that for every supported monomial α of degree d and every monomial δ of exactly that degree,

$$q_j - \alpha \cdot w + \delta \cdot w = q_k \quad \Longleftrightarrow \quad (j, \alpha) = (k, \delta).$$

All reversed exponents are nonnegative. This is verified against all 496 pure quadratic monomials, all 31 linear monomials, and the constant monomial: $45 \cdot 496 + 41 \cdot 31 + 8 = 23599$ interactions. The coefficient of a mixed product in Y_z^2 is two, while that of a square is one. The stated coefficient factors give exactly $2r_j$. All carrier exponents exceed Q_c. The supplied checker also reconstructs the 22 identities symbolically with the native ports left indeterminate, independently of the interaction count. □

Cross-degree collisions are harmless because a term of $A_1 Y_z$ is never multiplied by a quadratic monomial, and a term of $A_2 Y_z^2$ is never multiplied by a constant or a linear monomial. The three channels are fixed polynomials evaluated at the same base and the same single packed input. No extra existential coordinate is introduced.

4.1 A positive carrier valid before decoding

Let $b \geq 2$, $B = a_S b^2$, and

$$\ell = \sum_i B^{w_i}, \qquad y = g + \ell.$$

The shift by ℓ means that packed nonnegative digits n_i decode to the positive parent coordinates $n_i + 1$. By construction, the reversed exponents are distinct within each channel. Since each coefficient has magnitude below M,

$$|A_d(B)| < \frac{MB^{Q_c+1}}{B-1}.$$

Put $\varepsilon = M/(B-1) < 1/4$. The elementary inequality $y^2 + |y| + 1 \leq \frac{3}{2}(y^2+1)$ gives, for *every integer* y,

$$\tfrac{1}{2}B^{Q_c+1}(y^2+1) < \mathcal{F}(B,y) < 2B^{Q_c+1}(y^2+1). \tag{15}$$

In particular, positivity is available before a ratio bridge has recovered a bound on g. A carrier proved positive only on the eventual decoding range would not justify that proof order.

Define the source registers

$$N_0 = B^{W+1}, \qquad Z_c = B^{K_c+1}, \qquad A_T = \sum_j B^{q_j},$$

$$M_c = N_0 - 1 - (b-1)\ell, \qquad J_c = 2\mathcal{F}(B, g+\ell) + A_T,$$
$$S_c = g + N_0 J_c, \qquad T_c = M_c + N_0(B-2)A_T.$$

For every $g > 0$, $S_c > 0$. The base-B digits of M_c are $B - b$ at the 31 weights and $B - 1$ at the remaining positions through W, so $0 \leq M_c < N_0$ and $T_c \geq 0$, without any power-of-two assumption on b.

If $0 \leq g < 4bB^W$, then $g < N_0$, $\ell < 2B^W$, and $g + \ell < 5bB^W$. Equation (15) gives

$$0 < \mathcal{F}(B, g+\ell) < 52b^2B^{K_c}, \qquad 0 < J_c < 105b^2B^{K_c} < Z_c.$$

Also $(B-2)A_T < Z_c$. Thus $0 \leq S_c, T_c < B^{E_c}$. The stronger bound needed for the ten-integer adapter follows from

$$J_c + (B-2)A_T = 2\mathcal{F}(B, g+\ell) + (B-1)A_T < (104b^2+1)B^{K_c} < \tfrac{1}{4}Z_c.$$

Here $a_S > 420$. Since $g + M_c < 2N_0$ and $Z_c \geq 4$,

$$S_c + T_c < N_0(2 + Z_c/4) < N_0 Z_c = B^{E_c}. \tag{16}$$

4.2 Exact zero extraction with signed borrows

Assume b is a power of two. Then B and N_0 are powers of two. In the range just proved,

$$g \,\&\, M_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i B^{w_i}, \qquad 0 \leq n_i < b.$$

The tuple is unique. Put $z_i = n_i + 1$, so $1 \leq z_i \leq b$. If h_s is the coefficient of T^s in $\mathcal{F}(T, Y_z(T))$, the sum $z = \sum_i z_i \leq 31b$ gives

$$|h_s| \leq (M+1)z^2 + Mz + M + 1 < 2048Mb^2 = B/4.$$

The polynomial degree is at most K_c. Write the signed expansion

$$J_c = \sum_{s=0}^{K_c} (2h_s + \epsilon_s)B^s, \qquad \epsilon_s = \begin{cases} 1 & s \in \{q_1, \ldots, q_{22}\}, \\ 0 & \text{otherwise.} \end{cases}$$

Ordinary Euclidean digit normalization has incoming carry always zero or minus one, by induction from the strict coefficient bound. At a target, the normalized digit avoids the binary mask $B-2$ exactly when $h_s = 0$: a zero coefficient gives digit zero or one; a positive coefficient gives a digit at least two; a negative coefficient gives a normalized digit at least two after borrowing. This argument explicitly permits an incoming borrow.

Concatenation at the N_0 boundary and the extraction lemma prove

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad \begin{cases} g = \sum_i n_i B^{w_i}, \quad 0 \le n_i < b, \\ r_j(n_1+1, \ldots, n_{31}+1) = 0 \quad (1 \le j \le 22). \end{cases} \tag{17}$$

Bitwise AND is used only in the proof, never in the final equations.

For completeness, any positive parent solution gives nonnegative digits $n_i = z_i - 1$. They are not all zero: the fifth shifted residual equals one at the zero vector. Choose any sufficiently large power of two $b > \max_i n_i$. All weights are positive, so the resulting $g \ge B \ge b$, while $g < bB^W$, and every carry test passes. These witnesses exist at arbitrarily large admissible powers of two; the ten-integer construction may restrict them to $b = 4^s$.

5 Nine positive unknowns at degree 2,628,944

We compose the degree-separated source of Section 4 with an exact binary-population calibration and the inherited Jones ratio construction. The resulting equation has nine strictly positive unknowns, exact total degree 2 628 944, and 499 live binary arithmetic operations. Its separately translated nonnegative version has the same degree and 501 operations. The arithmetic proof below identifies the exact zero set; no intermediate register is an additional unknown.

5.1 Source registers and the binary threshold

Supply the nine positive coordinates

$$(\epsilon_+, g, h, s, w, \phi_+, i, j_+, n_+) \in \mathbb{Z}^9_{>0}, \qquad \phi = \phi_+ - 1, \quad j = j_+ - 1, \quad n = n_+ - 1.$$

Put $b = \epsilon_+ + 1$, $a_\star = 2^{1182968}$, and $B = a_\star b^2$. Use the source registers S_c, T_c from Section 4, including its raw positive word and positive carrier. Put $c_\star = 2^{9463734}$ and define

$$\begin{aligned} L &= 4bB^{2241}, & N &= c_\star bB^{1178}, \\ R &= (S_c + T_c)L + L - 1 - S_c, & G_{\mathrm{cap}} &= 32bB^{108}. \end{aligned} \tag{18}$$

All exponents are fixed nonnegative integers. In particular, N and the binary block length L are different polynomials.

The source theorem gives $S_c > 0$ and $T_c \ge 0$ on *every* outer assignment, before recovering a binary base or a bound on g. Hence

$$R = (L-1)S_c + LT_c + L - 1 \ge 8, \qquad N \ge 8, \qquad 0 < b \le R, N, \qquad G_{\mathrm{cap}} > 0.$$

If subsequently $8g < G_{\mathrm{cap}}$, then $g < 4bB^{108}$, and the source bounds give $S_c, T_c < B^{2241}$ and $S_c + T_c < L$.

Suppose also $b = 2^k$, and write $B = 2^v$, so $v = 1182968 + 2k$. The low mask occupies 109 base-B cells, with 31 allowed digit windows; the 22 output masks each contribute $v - 1$ bits. Thus

$$\mathrm{pop}(T_c) = 131v - 31k - 22, \qquad \log_2 L = 2 + k + 2241v.$$

The two disjoint binary blocks of R give the exact identity

$$\begin{aligned}\operatorname{pop}(R) &= \log_2 L + \operatorname{pop}(T_c) - \delta,\\ \delta &= \operatorname{pop}(S_c) + \operatorname{pop}(T_c) - \operatorname{pop}(S_c + T_c) \geq 0.\end{aligned}$$

Here $\delta = 0$ precisely when S_c and T_c have disjoint binary bits. Furthermore,

$$\begin{aligned}\log_2 L + \operatorname{pop}(T_c) &= 2372v - 30k - 20\\ &= 2\big(8 \cdot 1182968 - 10 + k + 1178v\big) = 2\log_2 N.\end{aligned}$$

Legendre's formula therefore proves

$$N^2 \mid \binom{2R}{R} \quad\Longleftrightarrow\quad S_c \,\&\, T_c = 0. \tag{19}$$

The binomial coefficient and bitwise symbol occur only in the proof. There is no leading L^2 term in R; its omission is justified by the unconditional positivity and exact block calculation above.

5.2 Complete polynomial definition

The following are polynomial registers. Letters reused from the source construction are local to this subsection:

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY,\\
P &= 2M_0^2 U, & K &= R + 1 + h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R + 1, & W_1 &= bw, & C &= B_1 + W_1 + \phi,\\
J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1), & &\\
D &= (A^2-1)C^2 + 1, & E &= iJDC^2, & &\\
F &= (A^2-1)E^2 + 1, & G &= A + F(CD + 1 - A), & &\\
H &= C + F(B_1 + jC), & I &= (G^2-1)H^2 + 1. & &
\end{aligned}$$

Set

$$\begin{aligned}
V_0 &= PK, & a_0 &= V_0^2 - K^2 + 1, & a_1 &= 64DFI,\\
\mu &= \big(K^2 - 4(C - KY)^2\big)G_{\mathrm{cap}} - 8gK^2, & & & &\\
t_0 &= iJC, & T_0 &= t_0(8D^2 - 4D - 1), & V_1 &= T_0 GH,\\
\Omega &= J\mu, & E_0 &= Jn + Q, & \Theta &= E_0 - \Omega(V_0 + V_1).
\end{aligned}$$

The promised equation is

$$\boxed{P_9 = \big(\Theta^2 - \Omega^2(a_0 + a_1)\big)^2 - 4\Omega^4 a_0 a_1 = 0.} \tag{20}$$

The factor 64 belongs to the present definition of a_1; the inherited ratio lemmas use the unscaled square predicate for DFI.

5.3 The ratio interface and an exact quotient change

The supplied `NineRatio.lean` and its accompanying mathematical proof give the following numerical contract. Under the unconditional preconditions established above, the four conditions

$$a_0 \text{ is a square}, \quad DFI \text{ is a square}, \quad J \mid Q, \quad \mu > 0 \tag{R}$$

recover b as a power of two, $8g < G_{\mathrm{cap}}$, and the binomial condition in (19). Conversely, the binomial condition, a binary b, and the stronger bound $32g < G_{\mathrm{cap}}$ supply the remaining ratio witnesses. The change from the original ratio variables to (R) is exact, as follows.

Originally $H_{\mathrm{old}} = B_1 + j_{\mathrm{old}}C$, with $j_{\mathrm{old}} > 0$, and the divisibility condition is

$$JF \mid QF + J(H_{\mathrm{old}} - C).$$

Since E is divisible by both J and C, one has $F \equiv 1 \pmod{J}$ and $F \equiv 1 \pmod{C}$. The displayed divisibility is equivalent to $J \mid Q$ and $F \mid H_{\mathrm{old}} - C$. Write $z = (H_{\mathrm{old}} - C)/F$. Then $z > 0$ and $z \equiv B_1 \pmod{C}$. Because $0 < B_1 < C$, there is a unique $j \geq 0$ with $z = B_1 + jC$. This gives exactly $H = C + F(B_1 + jC)$. Conversely,

$$j_{\mathrm{old}} = 1 + \frac{F-1}{C}B_1 + Fj > 0$$

is an integer and reconstructs the original H_{old}. The division occurs only in this proof of a witness bijection. Neither an added variable nor an auxiliary cofinality assertion is needed.

5.4 A polynomial majorant and recovery in the correct order

Every register needed below is positive, with $A \geq 2$, $J \geq 3$, $K > 1$, $D \geq A^2$, and $H > C > 1$. Also $Q > 0$. Define the polynomial

$$\mathcal{E} = t_0^2(8D+1) - 64D.$$

The identities $E = t_0DC$ and $(A^2-1)C^2 = D-1$ give

$$\begin{aligned} F &= t_0^2D^2(D-1) + 1, \\ T_0^2 - 64DF &= \mathcal{E} > 0, \\ V_1^2 - a_1 &= \mathcal{E}(GH)^2 + 64DF(H^2-1) > 0. \end{aligned} \tag{21}$$

Indeed $t_0 \geq 3$, so $\mathcal{E} \geq 9(8D+1) - 64D > 0$. Together with $V_0^2 - a_0 = K^2 - 1 > 0$, these prove

$$0 < \sqrt{a_0} < V_0, \qquad 0 < \sqrt{a_1} < V_1$$

without assuming either radicand is a square. Moreover $J > W_1$, $0 < Q/J < 3C$, and $V_1 > 3C$; hence $V_1 > Q/J$. For example, $U = N^2w \geq 2W_1$ implies $J > W_1$, while $T_0 > t_0 \geq 3C$ proves the latter majorant bound.

Lemma 5.1 (Exact finalizer)**.** *For fixed outer coordinates in the stated domain, there is an $n \geq 0$ satisfying (20) if and only if the four conditions (R) hold.*

Proof. If $\mu = 0$, then $\Omega = 0$ and $P_9 = (Jn+Q)^4 > 0$. Otherwise the real factorization of the two-root norm implies, for some signs, that

$$n + \frac{Q}{J} = \mu(V_0 + V_1 \pm \sqrt{a_0} \pm \sqrt{a_1}). \tag{$*$}$$

The parenthesized expression is positive by the strict majorants; the left side is positive because $n \geq 0$ and $Q > 0$. Thus $\mu > 0$ is obtained *before* any integrality claim about the radicals.

The margin inequality now gives $|C - KY| < K/2$, whence $C > K$ because $Y \geq 64$. Also $P < 2A^2$, $F \geq A^2$, and $I \geq 1$. Consequently

$$a_1 \geq 64(A^2-1)A^2C^2 \geq 48A^4C^2 > 4A^4K^2 > P^2K^2 > a_0.$$

Let $z = \Theta/\Omega$. The vanishing norm says $z = \pm\sqrt{a_0} \pm \sqrt{a_1}$. This rational number is nonzero because $a_1 > a_0$. Squaring after isolating either root gives, with a suitable sign,

$$2z(\pm\sqrt{a_0}) = z^2 + a_0 - a_1.$$

Both roots are therefore rational, hence integral. In particular DFI is a square, since a rational square root of an integer is integral. Equation (*) now gives $Q/J \in \mathbb{Z}$, proving (R).

Conversely, under (R), choose the positive integer square roots and set

$$n = \mu(V_0 + V_1 + \sqrt{a_0} + \sqrt{a_1}) - Q/J.$$

This is an integer and is positive, since $\mu \geq 1$ and $V_1 > Q/J$. One real factor of the norm vanishes, proving $P_9 = 0$. □

The ratio contract, the source theorem, and this lemma prove soundness. For completeness, encode a positive parent solution by digits $n_i = z_i - 1$. The shifted parent has no zero at the zero vector, so its code is positive. Choose any sufficiently large binary b exceeding all digits. The source then supplies $g < bB^{108}$, hence $32g < G_{\mathrm{cap}}$, and the binomial condition. Apply ratio completeness, the exact quotient change, and the lemma. Composing with the pinned compiler and K215 calibration proves

$$\mathrm{ZFC} \vdash \ (\neg\exists \mathbf{x} \in \mathbb{Z}^9_{>0}\ P_9(\mathbf{x}) = 0) \iff \mathrm{Con}(\mathrm{ZFC}).$$

For the nine-natural version use independent coordinates $\epsilon, g_0, h_0, s_0, w_0, \phi, i_0, j, n \in \mathbb{N}$, put $b = \epsilon + 2$, and replace g, h, s, w, i by $g_0 + 1, h_0 + 1, s_0 + 1, w_0 + 1, i_0 + 1$. These are exact domain translations; the positive polynomial is not silently reinterpreted over unrestricted integers.

5.5 Exact degree and evaluation cost

The apparent top degree of the norm cancels. The supplied degree-certificate circuit resolves that cancellation using (21). Write $\Delta_1 = V_1^2 - a_1$. The exact polynomial identity is

$$\begin{aligned}\Theta^2 - \Omega^2(a_0 + a_1) = E_0^2 - 2E_0\Omega(V_0 + V_1)& \\ + \Omega^2(K^2 - 1 + \Delta_1 + 2V_0V_1),& \\ \Delta_1 = \mathcal{E}(GH)^2 + 64DF(H^2 - 1).&\end{aligned}$$

Let $\rho = \deg R = 8963$ and $\nu = \deg N = 2357$. Propagation of total-degree upper bounds through these identities gives

$$\deg P_9 = 188\rho + 400\nu + 8W + 236 = 2\,628\,944, \qquad W = 108.$$

For clarity, $\deg \Omega = 92\,184$, $\deg \Delta_1 = 1\,130\,104$, and $\Omega^2\Delta_1$ uniquely supplies degree $1\,314\,472$ in the inner expression. Its square has degree $2\,628\,944$; the other outer term has degree $1\,791\,262$.

The upper bound is attained. Set every supplied positive coordinate equal to a single indeterminate x. The leading coefficients of R and N are respectively $8a_\star^{4481}$ and $c_\star a_\star^{1178}$. The leading coefficient of the final polynomial is exactly

$$\boxed{2^{742}a_\star^{1314060}c_\star^{400} > 0.}$$

This proves exact total degree, rather than a syntactic bound or a random numerical estimate. The same coefficient survives the natural-coordinate translations. The primary circuit has 170 additions, 25 subtractions, and 304 multiplications, totaling 499 operations; the natural version has 175 additions, 22 subtractions, and the same 304 multiplications, totaling 501. Fixed-power evaluation is expanded into these binary gates. The larger cancellation-resolved circuit is a degree certificate and is not substituted for the primary evaluator's cost.

6 Ten unrestricted integers

Supply exactly ten unrestricted integer coordinates

$$(f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}.$$

Use the source of Section 4, with

$$b = 1 + 3f, \qquad B = a_S b^2, \qquad \mathcal{C} = bB^{108}, \qquad L_c = B^{2241},$$
$$X = L_c^2 + (S_c + T_c)L_c + L_c - 1 - S_c,$$
$$Y = a_S^{15}\, 2^{\kappa-21} B^{2356} b.$$

The monomial cap is $\mathcal{C} = a_S^{108} b^{217}$, as required by the inherited Sun bridge. The source output Y is a fixed polynomial, not an exponential with a variable exponent.

6.1 The arithmetic source contract

Before recovering a binary base, the reverse bridge has $f \neq 0$, b an integer square, and $0 \leq g < 2\mathcal{C}$. Since $b \equiv 1 \pmod 3$, these hypotheses imply $b \geq 4$ and $f > 0$. The preceding source proof applies and gives $S_c, T_c \geq 0$ and $S_c + T_c < L_c$, including the stronger summed bound (16). Hence

$$X \geq L_c^2 > 3b, \qquad Y > \max(b, 256).$$

The exponent κ is even, so $B \equiv N_0 \equiv L_c \equiv 1 \pmod 3$. We have $M_c \equiv 0$, $A_T \equiv 22$, and $T_c \equiv -22 \pmod 3$. Consequently

$$X \equiv 1 + T_c \equiv 0 \pmod 3.$$

These range and congruence facts do not assume b is a power of two.

After binary-base recovery, write $b = 2^s$, $B = 2^v$. For nonnegative S, T with $S + T < L_c$, the three disjoint blocks in $L_c^2 + (S+T)L_c + L_c - 1 - S$ give

$$\operatorname{pop}(X) = 1 + \log_2 L_c + \operatorname{pop}(T_c) - \big(\operatorname{pop}(S_c) + \operatorname{pop}(T_c) - \operatorname{pop}(S_c + T_c)\big).$$

The subtracted carry deficit is nonnegative, and vanishes exactly when $S_c \,\&\, T_c = 0$. The exact mask population is $\operatorname{pop}(T_c) = 131v - 31s - 22$, so

$$1 + \log_2 L_c + \operatorname{pop}(T_c) = 2372v - 31s - 21 = \log_2 Y.$$

Legendre's formula therefore proves

$$Y \mid \binom{2X}{X} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0.$$

Conversely a parent zero can be encoded with arbitrarily large $b = 4^s$, putting $f = (b-1)/3$. It has $b \leq g < \mathcal{C}$ and the required divisibility. This is the full source contract; the Sun bridge does not require Y to be a square or equal to L_c^2.

6.2 Literal final polynomial

The following finalizer is inherited from the inspected package-31 and wave-7 construction; only its source polynomials have changed. Define

$$L = lY, \quad U = 2LX, \quad V = 4gwY, \quad W_b = bw,$$
$$A = U(V+1), \quad B_* = 2X+1, \quad C = B_* + (A-2)h,$$
$$P = U^2V, \quad K = X + 1 + k(P-2), \quad S_* = 2A-5,$$
$$T_* = 3W_bC - 2(W_b^2-1), \quad R_2 = (P^2-4)K^2 + 4,$$
$$\Delta_* = 8\mathcal{C}^3gK^2 - g^2\{32(C-KL)^2\mathcal{C}^3 + g^2K^2\},$$
$$O = f^2l^2\Delta_*, \quad \xi = zx, \quad D = (A^2-4)C^2 + 4,$$
$$E = C^2D\xi, \quad F = 4(A^2-4)E^2 + 1,$$
$$G_* = 2(A+2)(A-2)^2E^2 - 1, \quad \Omega_* = 16E,$$
$$H_* = C + (B_*-C)F + \Omega_*Fy, \quad I_* = (G_*^2-1)H_*^2 + 1.$$

Put

$$r = T_* + S_*y, \qquad \Sigma = \xi S_*^5,$$
$$\Theta_* = (r^3 - rS_*^2 - S_*^3)(r^2 + 23S_*^2),$$
$$a_0 = b\{1 + (3O-4)b^2z^2\}, \qquad a_1 = R_2, \qquad a_2 = DFI_*,$$
$$w_0 = 1, \qquad w_1 = a_0, \qquad w_2 = a_0(P^2 + K^2 + 3),$$
$$Z_* = \Sigma m - \Theta_*, \qquad A_i = \Sigma^2 w_i^2 a_i \quad (0 \le i \le 2),$$
$$V_* = Z_*^2 + A_2 - A_0 - A_1, \qquad W_* = V_*^2 + 4Z_*^2A_2 - 4A_0A_1.$$

The single equation is

$$\boxed{F_{10} = W_*^2 - 16V_*^2Z_*^2A_2 = 0.} \tag{22}$$

All powers in these formulas are fixed nonnegative integers. In particular, no square root, factorial, binomial coefficient, bit operation, or divisibility predicate occurs in the actual polynomial.

6.3 Why the inherited finalizer still applies

Sun's arithmetic bridge, specialized as in the preserved proof, supplies both directions for any source satisfying the contract above. The reverse direction requires b square, $O > 0$, $\xi \neq 0$, R_2 and DFI_* square, and $S_* \mid T_*$. It recovers b as a power of two, $0 < g < 2\mathcal{C}$, and the central-binomial divisibility. The forward direction supplies these conditions, together with the residue interpolation needed for the finalizer. These are the same arithmetic bridge hypotheses as in the inspected construction.

For clarity, the finalizer's main logical steps are recorded here. The polynomial $(v^3 - v - 1)(v^2 + 23)$ has a root modulo every positive integer and no rational root. Prime valuations in its homogeneous factors show that $\Sigma \mid \Theta_*$ implies $S_* \mid T_*$. The parity of S_* makes it nonzero, and the absence of rational roots makes $\Theta_* \neq 0$. If $\Sigma = 0$, then $F_{10} = \Theta_*^8 \neq 0$, excluding that branch.

For $z \neq 0$, the factors b and $1 + (3O-4)b^2z^2$ are coprime. If their product is a square, they are signed squares; $b \equiv 1 \pmod 3$ excludes a negative square and the zero-factor exception. The positivity Pell lemma then gives $O > 0$. Conversely, the forward bridge has $O \ge 4$; a suitable positive Pell unit makes a_0 a square. Residue interpolation supplies a multiple $\xi = zx$ and the required class of y, making a_2 a square and $\Sigma \mid \Theta_*$.

The structural weights satisfy

$$|w_1| > \sqrt{|a_0|}, \qquad |w_2| > \sqrt{|a_0|} + |w_1|\sqrt{|a_1|}.$$

Here $|a_0| > 1$ whenever $z \neq 0$, and $\sqrt{|a_1|} \leq P^2 + K^2 + 2$. In the multiquadratic field, any nontrivial sign-changing automorphism of a weighted root sum has a largest changed index whose magnitude exceeds the sum of all smaller changed magnitudes. Thus a rational weighted sum forces every radicand to be an integer square. The exact identity

$$F_{10} = \prod_{\epsilon \in \{\pm 1\}^3} \left(Z_* + \sum_{i=0}^{2} \epsilon_i \sqrt{A_i} \right)$$

now proves the finalizer equivalence. This argument also handles negative radicands using complex absolute values, and zero radicands separately. At a zero, the bridge and (17) recover a parent solution. From a parent solution, choose the forward bridge witnesses and set m to cancel one integral signed factor.

The full inherited interpolation, local-root, and bridge proof is kept in `upstream/wave7_sun_popcount.tex`, with its precise primary references [3, 4]. The root verifier checks the norm identity by exact polynomial multiplication; the unbounded bridge is an identified mathematical dependency.

6.4 Exact degree and circuit cost

The source registers have degrees

$$\deg X = 8964, \qquad \deg Y = 4713, \qquad \deg \mathcal{C} = 217.$$

Representative downstream degrees are

	A	C	P	K	D	E	F
deg	18393	18394	32071	32072	73574	110364	257514

and

	G_*	H_*	I_*	Σ	Θ_*
deg	275907	367879	1287572	91967	91980.

The exact leading-coefficient certificate in `generated/ten_integer_degree.json` propagates the degree bound through every gate and exhibits a strictly positive final diagonal leader. The independent modular leading residues in Section 8 verify it with the actual fixed coefficients. The resulting exact degree is 8 317 400. There are 544 live binary operations: 174 additions, 26 subtractions, and 344 multiplications. The baseline comparison concerns unrestricted integer zeros throughout.

7 A quartic in forty-five unrestricted integers

The independent low-degree contribution reduces the inspected fifty-integer quartic to **forty-five unrestricted integer unknowns**, while retaining degree four. It uses 28 quadratic residuals instead of 29. Its expanded representation has 407 symbolic terms, compared with 362 in the chosen parent; the improvement concerns variable count at fixed degree, not monomial count or evaluation cost.

This construction keeps the complete package-34 coefficient tuple

`mixed-parity-unrounded-1064921`,

with $B = 2^{1064921}$, $\beta = B - 1$, and ports β, K_0, M_C, M_F. The last port is the already shifted $M_{F,\text{native}} + B - 1$. It is a separate complete binding from the tuple used for the principal twelve- and thirteen-variable equations. The four raw payloads, their finite construction, the literal K215 transition table, and the relevant inherited proofs are included in `quartic/upstream/`. The canonical accepting witnesses for this tuple have $q = 2^t$ with t odd.

7.1 The two guard reductions

Write

$$\mathrm{Pos}(u, v, t) = 1 + u^2 + v^2 + t(t + 1).$$

This expression is positive on all integer triples and represents every positive integer. For the converse, apply the three-square theorem to $4n - 3$: its residue modulo four forces one odd square and two even squares, yielding the displayed representation after division by four.

Lemma 7.1 (A common-sign guard)**.** *Suppose $q > 0$ and $C = q - F - \alpha$. Then*

$$F\alpha > 0 \quad \textit{and} \quad FC > 0 \quad \Longleftrightarrow \quad F, \alpha, C > 0.$$

Proof. The two products make the three nonzero integers have the same sign. Their sum is $q > 0$, so that sign is positive. The converse is immediate. □

Lemma 7.2 (A dyadic guard)**.** *For $\beta \geq 3$, the equation $\beta J + 1 = 2v^2$, with $J, v \in \mathbb{Z}$, implies $J > 0$. Every canonical radix $q = 2^t$ with t odd satisfies this equation with $v = 2^{(t-1)/2}$.*

Proof. If $J \leq -1$, its left side is negative; if $J = 0$, it is one, which cannot be twice an integer square. The second assertion is a direct substitution. □

7.2 The complete quadratic system

The 32 arithmetic coordinates are

$$J, F, \alpha, z, f, h, r, v_s, w, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta, u.$$

Add one coordinate v_q and four triples $\boldsymbol{e}, \boldsymbol{\xi}, \boldsymbol{b}, \boldsymbol{d}$. All 45 coordinates range over $\mathbb{Z}$. In the following formulas, the letters

$$q = \beta J + 1, \quad C = q - F - \alpha, \quad k = \eta + \zeta, \quad H = 4a + 3,$$
$$R = (q - F)(N - q) - CL + (M_C + qM_F)J$$

are polynomial abbreviations, not additional coordinates. Retain the following 23 arithmetic equations:

$$\begin{aligned}
L &= q^2 - 1, & N &= q(L + 1),\\
X &= wq, & a &= E + (2v_s + 1)N,\\
E &= X(a - E), & c &= k(a - E) + \eta,\\
P &= E(c - \eta) + k, & \tau^2 &= P(P - k) + 1,\\
D &= X + ac + \gamma H, & \Delta &= a^2 + H,\\
C_2 &= c^2, & D^2 &= \Delta C_2 + 1,\\
k &= R + 1 + hE, & (K_0 + w)C + q - F - z(q - 1) &= 1,\\
F_2 &= f^2, & V &= of - c,\\
V - cj + k - 1 - hE &= 0, & g &= y^2 - V^2,\\
I &= (2r + 1)C_2, & B_{\mathrm{aux}} &= I^2,\\
B_{\mathrm{aux}} &= \Delta(F_2 - 1), & y^2 - B_{\mathrm{aux}}g &= 1,\\
w &= u^2.
\end{aligned}$$

Add the five quadratic guards

$$\eta = \mathrm{Pos}(\boldsymbol{e}), \qquad \zeta = \mathrm{Pos}(\boldsymbol{\xi}), \qquad q = 2v_q^2,$$
$$F\alpha = \mathrm{Pos}(\boldsymbol{b}), \qquad FC = \mathrm{Pos}(\boldsymbol{d}).$$

Here $\mathrm{Pos}(\boldsymbol{e})$ means evaluation at its three coordinates. Every displayed residual is quadratic. If they are $r_1, \ldots, r_{28}$, define the single equation

$$\boxed{Q_{45} = \sum_{i=1}^{28} r_i^2 = 0.} \tag{23}$$

The coefficient of τ^4 is one, so the degree is exactly four. The literal ordered residuals, variable list, and exact expansion are `quartic/quartic45.json` and `quartic/quartic45_expanded.json`.

7.3 Why the unguarded transport quotient is harmless

The old fifty-variable system had independent ternary guards for $J, F, \alpha, z, \eta, \zeta$. Delete the first four guards and their twelve private variables; retain the last two; add v_q and the two new triples. Thus the variable count is $50 - 12 + 7 = 45$, and the residual count is $29 - 4 + 3 = 28$. Because these are domain changes, an algebraic identity alone would not prove the required existential equivalence.

At a zero of (23), the dyadic guard gives $J > 0$, and the common-sign guard gives $F, \alpha, C > 0$. The retained guards give $\eta, \zeta > 0$. The only missing original positivity condition is $z > 0$. Its recovery must respect the proof order.

In the inherited signed Pell proof, the early use of $z > 0$ was solely to infer $C \geq 0$ from transport. We already have the stronger conclusion $C > 0$, independently. Hence the same packing bounds follow:

$$q(q^2 - 1) < R < q^4 - 2, \qquad R - 2 > 3q + 1.$$

The square-scale exclusion gives $w > 0$. The existing odd-scale argument, with $s = 2v_s + 1$, excludes negative s; it uses the Pell equations, these bounds, and the exposed congruences $V \equiv -c \pmod{|f|}$, $V \equiv -R \pmod{|c|}$, without another use of z. Positive-scale rank recovery and the signed main-root argument then give

$$X = 2^R.$$

For a detailed dependency trace see the included `ODD_SCALE_27_PROOF.md`, Sections 1–4, and `SIGNED_DOMAIN_29.md`, through its signed-gamma and ordinary-strong arguments. This explicitly supplies $C > 0$ in place of the original preliminary transport inference; it does not invoke the final signed-domain theorem before restoring its guard.

Since $R > 3q$, we have $w = 2^R/q > q > F$. The retained transport equation now reads

$$(K_0 + w)C = (z - 1)(q - 1) + F.$$

Its left side exceeds F, so $z > 1$. All original positivity conditions are therefore restored. Choosing their ternary representations produces a zero of the old fifty-variable quartic, and its inherited compiler theorem yields a finite ZFC refutation.

Conversely, canonical accepting witnesses have $q = 2^t$ with odd t, and $F, \alpha, C > 0$. Set $v_q = 2^{(t-1)/2}$ and use the positive representations of $F\alpha$ and FC. Every new equation holds. Consequently the ordinary mathematical argument gives, formalizably in ZFC,

$$\text{there is no integer zero of } Q_{45} \quad \Longleftrightarrow \quad \mathrm{Con}(\mathrm{ZFC}).$$

The portable finite audit checks all 28 residual degrees, the 45 live coordinates, the exact expansion, $[\tau^4]Q_{45} = 1$, the complete fixed payloads, and the pinned inherited bytes. The unbounded Pell and compiler arguments remain explicit mathematical dependencies; this section does not claim a new end-to-end Lean/Rocq certificate or an exported object-language ZFC derivation.

8 Verification, reproduction, and formalization status

8.1 What the finite audits establish

The new proofs separate unbounded implications from finite certificates. The source checker reconstructs all 22 target identities as polynomial identities with four indeterminate coefficient ports, and exhausts all 23,599 relevant source/expansion interactions. The toy source checker deliberately allows a linear and a quadratic term to share a reversed position. It checks 1,020 source/carry cases, including three genuine zero cases, 680 forbidden-digit controls, and 36 unconditional-positivity regressions. These diagnostics would be meaningless as a proof of ZFC consistency; their purpose is to expose implementation mistakes in the source transformation.

The root verifier reads the emitted circuits without importing their builder. It authenticates all native and derived coefficient bytes, checks every gate reference, all variable leaves, liveness, and operation counts. It compares the complete output and key named registers with independent literal formulas at 72 deterministic modular assignments per equation. Four formal polynomial identities check the Pell majorant, its defect, the nine-variable degree cancellation, and the eight-factor three-radical norm. The latter norm has 35 terms.

The exact-degree check propagates a multivariate upper bound and the coefficient at that degree on a diagonal. The cancellation-resolved identity is used for nine variables. With the actual coefficient payloads, the independently obtained leading residues are

equation	101	1009	1000003
F_9	77	467	509827
F_{10}	80	969	99517

Any nonzero residue proves the required lower bound. Thus the reported degrees are exact, not syntactic upper bounds or estimates from sampled function values. The low-degree checker evaluates full univariate polynomials with the actual tuple, using $t = 3x$ and all other coordinates equal to x to avoid a vanishing top form. The universal upper bounds are supplied by the cancellation identities in its proof. The quartic checker expands the sum of squares exactly and verifies its degree-four coefficient, all 45 live coordinates, 28 residuals, and the separate coefficient binding.

8.2 Reproduction commands

All computational audits use the Python standard library and can be run from the packet root:

```
python3 build.py
python3 materialize.py
python3 verify.py
python3 data/verify_channels.py
python3 data/check_toy_channels.py
python3 coefficients/audit_coefficients.py
python3 low_degree/verify.py
python3 quartic/build_verify.py
```

The all-in-one runner `reproduce.py` also checks the pinned local sources and assembles a machine-readable receipt. The low-degree builders and both coefficient builders are supplied for regeneration. The source position search is not required: the small fixed certificate is the proof input. The two source checkers explicitly reject Python optimization flags, because their internal assertions are part of their audit.

Compile the paper from `report/` with two passes of `pdflatex-halt-on-errormain.tex`. The final packet includes the compiled PDF and all its TeX inputs. `MANIFEST.sha256` records the distributed files. An extracted copy was used for the final reproduction check, so the executable audits do not depend on the development checkout.

8.3 Ordinary provability versus an exported object-language proof

The intended consistency predicate is syntactic: there is no finite derivation of falsity from the recursively presented axioms of ZFC. The accepted K215 proof states that its blank halting is equivalent to the existence of such a derivation, by ordinary reasoning formalizable in ZFC. The present work composes that inherited theorem with explicit new polynomial reductions. Integer arithmetic, finite coefficient extraction, Pell recurrences, exact divisibility, and the witness constructions used here can all be developed internally in ZFC; none requires assuming Con(ZFC).

This mathematical claim must be distinguished from the repository's remaining proof-assistant obligation. At the inspected revision, `Computability/DiophantineConsistency/OBJECT_ZFC.md` states that an external Lean theorem about `BProvZFCax_s[]Form.fBot` is not itself an object-language derivation of the equivalence. The concrete target sentences and a complete derivation of their equivalence are not yet assembled. A formal proof must handle the model's own arithmetic, including nonstandard witnesses, and cannot use numeral agreement or bounded consistency as a substitute. We preserve this boundary: the archive supplies new mathematical proofs and exact executable finite certificates, not a new kernel-checked end-to-end theorem or a completed `BProv` endpoint.

Neither the existence nor the absence of solutions is asserted unconditionally in this paper. The equivalences are the result.

A Complete channel polynomials

The following table specifies every term of the three channel polynomials used in Section 4. A row (d, e, c) contributes cT^e to $A_d(T)$. The four symbols in its coefficient column are the fixed native tuple of Section 2. Thus the table, the fixed payloads, the weights and the displayed register formulas give an explicit polynomial without an appeal to a general existence theorem for Diophantine representations.

Degree channel d	Exponent e	Coefficient c
0	39	-2
0	238	$2K_0$
0	368	-2
0	1093	-6
0	1147	-2
0	1388	-2
0	1621	2
0	1861	-2
1	0	2
1	29	2
1	32	2
1	134	$2\beta + 2\beta K_0$

Degree channel d	Exponent e	Coefficient c
1	140	2
1	159	$-2K_0$
1	162	$-2-2K_0$
1	238	2
1	461	2
1	499	2
1	539	-2
1	545	2
1	652	2
1	732	2
1	741	-2
1	760	2
1	828	2
1	966	2
1	977	-4β
1	984	2
1	1001	2
1	1090	-8
1	1251	2
1	1268	-2
1	1282	2
1	1284	-2β
1	1291	-2
1	1298	2
1	1362	2
1	1397	2
1	1404	-2
1	1407	-2
1	1515	-2
1	1517	$-2M_F-2M_C+4\beta$
1	1524	2
1	1545	-2
1	1582	-2
1	1671	2
1	1877	2
1	1882	-6
1	1910	-2
2	2	-1
2	9	-1
2	36	β
2	61	-1
2	64	-1
2	74	$-\beta$
2	111	-2
2	280	2
2	283	-1
2	337	-1
2	514	1
2	517	1
2	533	-1
2	536	-1
2	543	-2
2	603	-1
2	819	-2
2	824	1
2	843	-1

Degree channel d	Exponent e	Coefficient c
2	864	-1
2	873	$-2\beta^2$
2	1087	-2
2	1113	-2
2	1120	1
2	1123	1
2	1125	2
2	1144	-1
2	1164	$-\beta$
2	1187	$-\beta$
2	1385	1
2	1391	-1
2	1411	$-\beta$
2	1413	$-2\beta M_F + 2\beta^2$
2	1420	β
2	1439	1
2	1441	$-\beta$
2	1445	-1
2	1448	-1
2	1584	1
2	1623	-2
2	1633	2
2	1680	-1
2	1787	2
2	1879	-4
2	1910	-1

References

[1] V. Reshetnikov and contributors, *ProveIt*, inspected revision `4b4349064a750ba6347293b55874a7f92abdd0db`, especially `Computability/DiophantineConsistency/con-zfc-diophantine-forms` and `docs/incoming`. https://github.com/VladimirReshetnikov/ProveIt/tree/4b4349064a750ba6347293b55874a7f92abdd0db.

[2] J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47** (1982), 549–571. The numerical ratio interface used here is preserved in `upstream/NineRatio.lean`, with its supporting `Ratio.lean` and `ProductPell.lean` source.

[3] Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Science China Mathematics **64** (2021), 281–306. https://arxiv.org/html/1704.03504v7.

[4] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, 2026 preprint, version 2. The inherited ten-integer bridge is specialized from Propositions 6.1–6.2. https://arxiv.org/html/2609.01594v2.

[5] J. Bayer, M. David, M. Hassler, Yu. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, 2025 preprint. https://arxiv.org/html/2506.20909v2.